FOMC TONE ANALYSIS

Predicting Monetary Policy Stance from FOMC Minutes

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CS 5100 / August 12, 2026

RESEARCH QUESTION

Can public FOMC minutes be converted into a measurable hawkish-dovish policy stance index?

PROJECT CONTRIBUTIONS

- Reproducible NLP pipeline
- Independent human audit
- Exploratory market comparison

Pipeline and Models

01
Data
147 FOMC minutes
DGS10 / DGS2 yields

→

02
Preprocess
Clean HTML
Split into sentences

→

03
Models
Dictionary baseline
Weak-label FinBERT

→

04
Evidence
30-sentence audit
Treasury comparison

PRINCIPAL DICTIONARY-DERIVED TONE INDEX

100 × (hawkish − dovish) / all sentences → three-meeting moving average

Independent Classifier Evaluation

DICTIONARY

53.3%

16/30 correct
Macro-F1 0.531

WEAKLY SUPERVISED FINBERT

50.0%

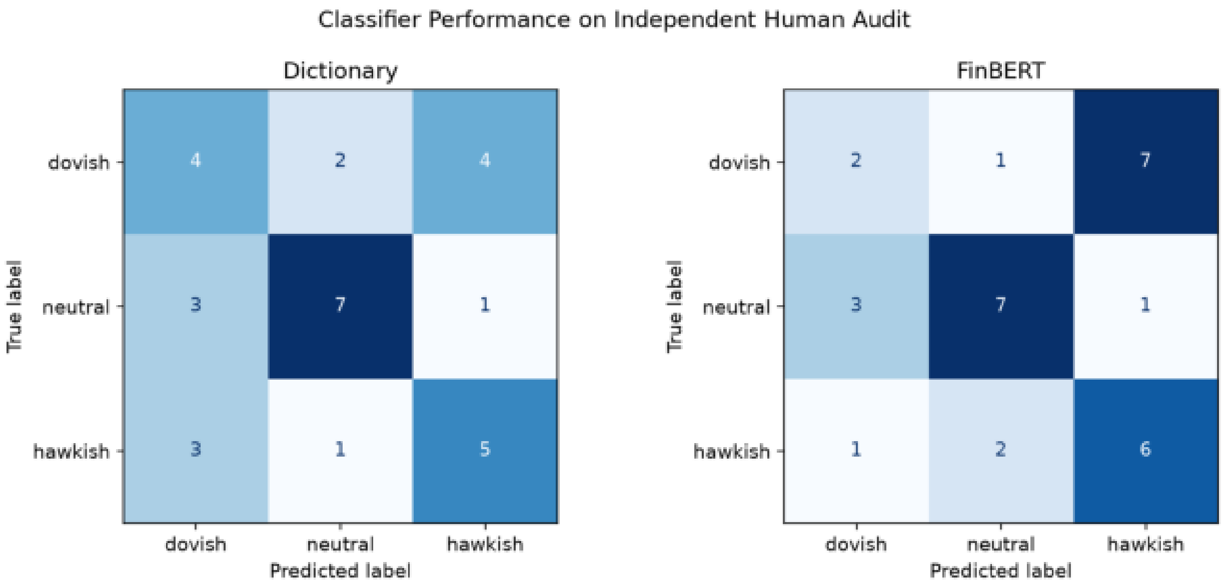
15/30 correct
Macro-F1 0.479

TAKEAWAY

One prediction separates the models—too little to establish meaningful superiority.

Both models beat the 33.3% chance baseline for three balanced classes, but not by a wide margin.
The small, single-annotator audit is preliminary; weak supervision may reproduce the dictionary's assumptions.

Audit Confusion Matrices

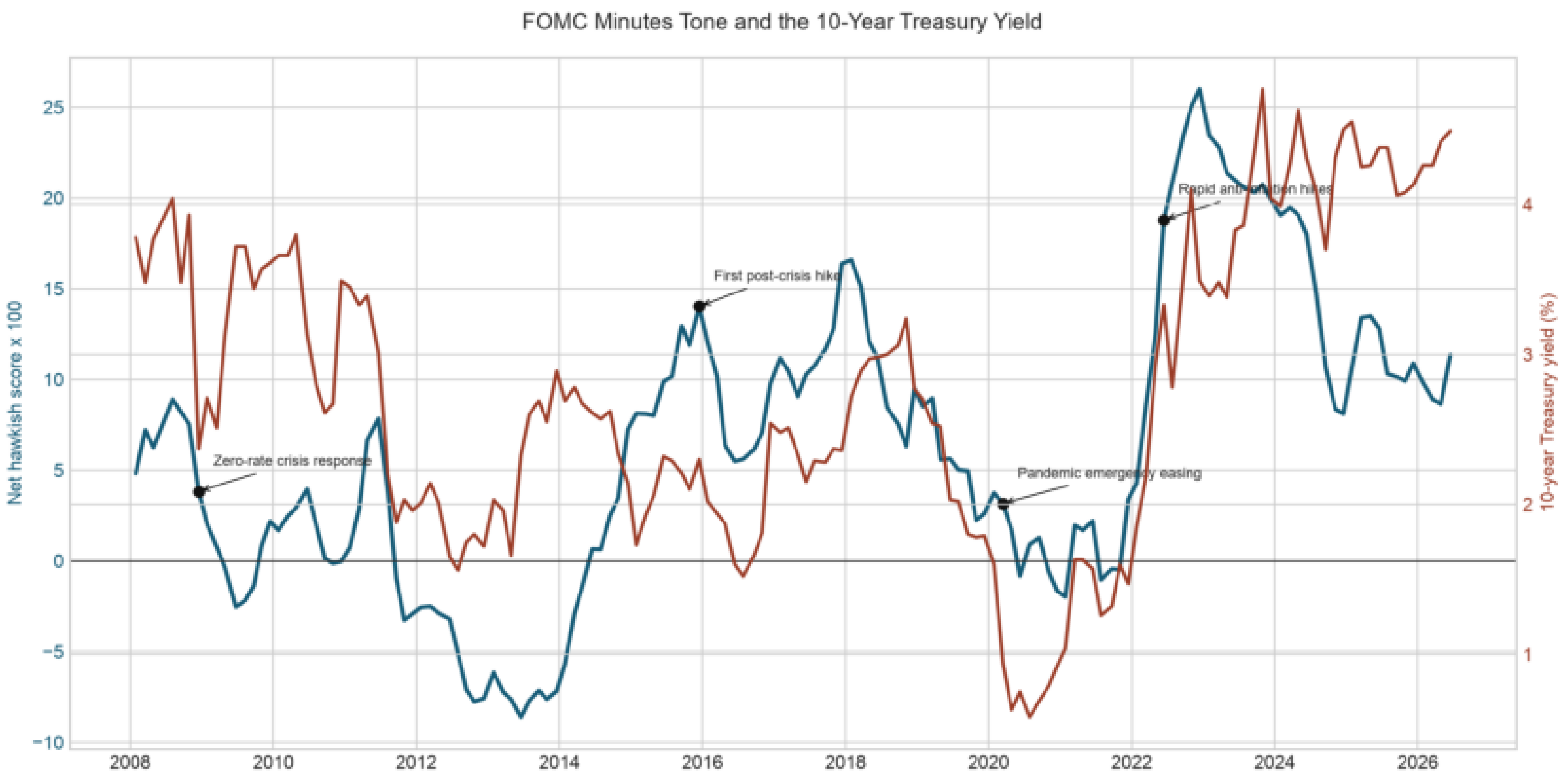


HAWKISH (2014-07-30): “Inflation firmed in recent months, and most participants anticipated that it would continue to move up toward the Committee's 2 percent objective.”

DOVISH (2010-06-23): “…existing margins of economic slack, although the anticipated decline in the unemployment rate was somewhat slower than in the previous projection.”

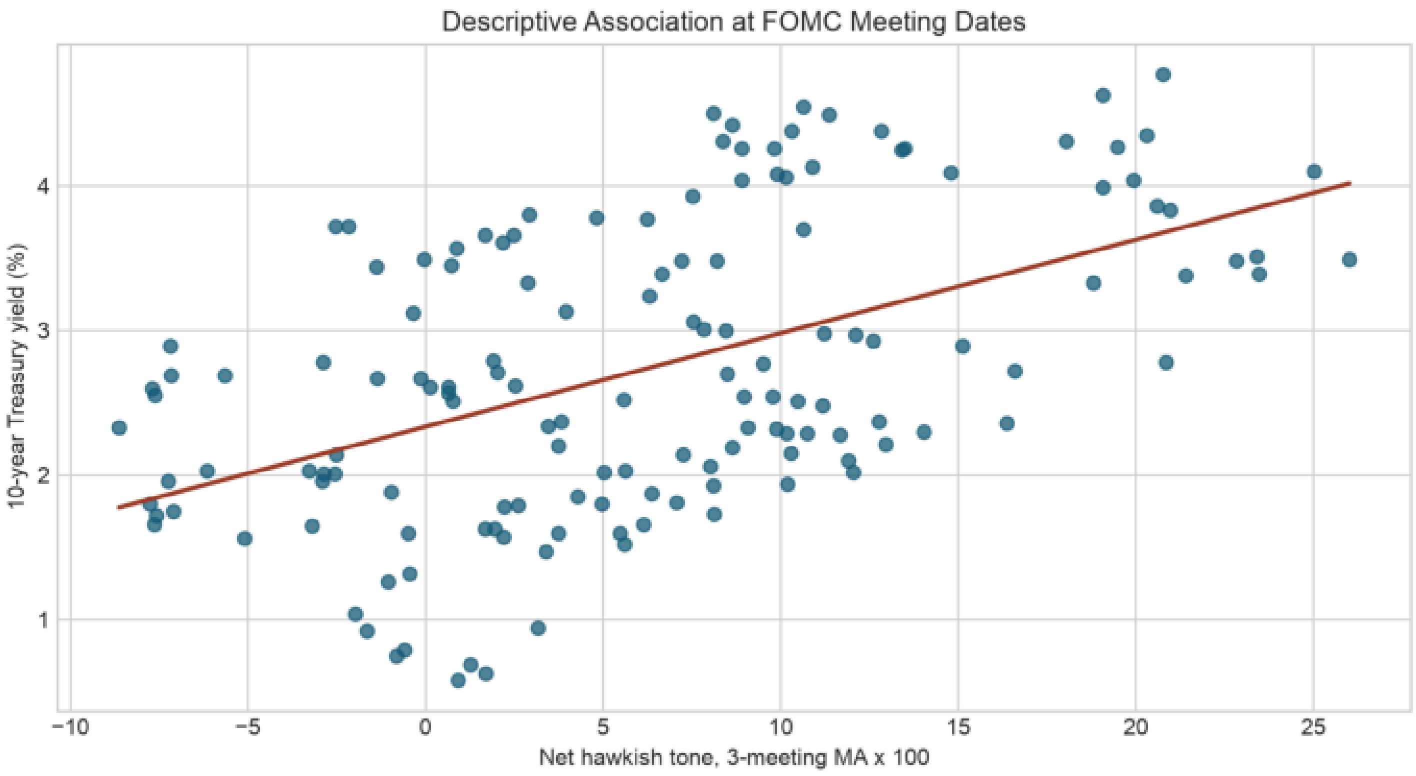
Independent human labels provide classifier ground truth. Both models were evaluated on the same 30 audited sentences.

Tone Across Policy Regimes



The three-meeting-smoothed, dictionary-derived tone index shifts around crisis, normalization, pandemic, and tightening regimes. The red series is the 10-year yield, compared directly on the next slide.

Tone and the 10-Year Treasury Yield



Tone and the 10-year yield move together across broad policy regimes. The association is descriptive; the full caveats follow on the next slide.

Sensitivity Across Yields and Lags

- Two yields × four tone lags were screened. The largest observed correlation was DGS2 at lag 2: $r = 0.806$ ($n = 145$).
- The maximum reflects specification selection across eight yield-and-lag combinations.
- Overlapping smoothing, persistence, and common policy-regime trends may inflate the association.

Not classifier validation, a causal result, or a minutes-release event study.

Conclusion

- The project delivers an end-to-end, reproducible monetary-policy text-classification workflow.
- On the 30-sentence audit, the dictionary made 16 correct predictions versus FinBERT's 15; this cannot establish a meaningful difference.
- Independent labels are essential when a comparison model learns from weak labels generated by the baseline.

Classifier accuracy, teacher-label agreement, and market association are different evidence.